

Platform Organization & Access Control Architecture

This document details the roles hierarchy, security gates, data isolation model, and user access matrix implemented across the DMX Academy platform. ---

1. User Roles & Hierarchy

The platform defines five distinct roles in the `Role` enum to support multi-tenant institute scoping and global content delivery.

Super Admin (`ADMIN`)

- **Scope:** Global (Platform-wide).
- **Key Function:** Manages core platform systems, creates global contests, designs coding problems, sets global AI configurations, and provisions new institutes and institute admins.

Institute Admin (`INSTITUTE_ADMIN`)

- **Scope:** Institute-bound.
- **Key Function:** Manages the local institute infrastructure, creates and assigns Batches/Cohorts, registers mentors, registers students, and creates institute-scoped contests.

Batch Manager (`BATCH_MANAGER`)

- **Scope:** Institute-bound.
- **Key Function:** Coordinates operations for specific target cohorts/batches, oversees student progress, uploads custom prep materials, and helps coordinate live instruction sessions.

Mentor (`MENTOR`)

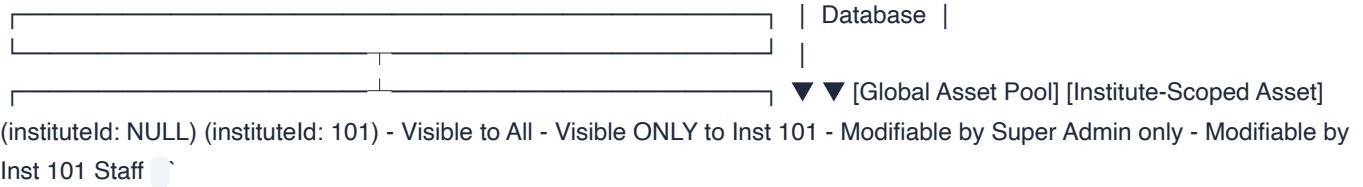
- **Scope:** Institute-bound.
- **Key Function:** Hosts interactive live classes, generates practice questions from uploaded PDFs, provides feedback on student performance, and sets programming contests.

Student (`USER`)

- **Scope:** Global or Institute-bound.
- **Key Function:** Attends classes, completes practice problems, competes in contests, and participates in AI-driven Viva voice reviews.

2. Data Scoping & Multi-Tenancy Isolation

Data isolation is built directly into the database layers (`schema.prisma`) using the `instituteId` field:



Global Scope (`instituteId: null`)

- Asset examples: Global Problems, Global Contests, Global Study Materials, Global Question Bank.
- These assets are created by the Super Admin to provide standard core curricula and public practice materials available to everyone on the platform.

Institute Scope (`instituteId: Int`)

- Asset examples: Private Problems, Target Batch Contests, Cohort Study Materials, Local Viva Questions.
- These assets are bound to a specific institute ID. They are created by institute staff and are completely hidden from all other competitor institutes.

3. Feature Access Matrix

The following matrix describes what each role can do across the different platform features.

Feature Area	Super Admin (<code>ADMIN</code>)	Institute Admin (<code>INSTITUTE_ADMIN</code>)	Batch Manager (<code>BATCH_MANAGER</code>)	Mentor (<code>MENTOR</code>)	Global Student (<code>USER</code>)	Institute Student (<code>USER</code>)
AI Viva Settings	Full Control (Read/Write)	Denied	Denied	Denied	Full Control (Read/Write)	Denied
Global Problems	Full Control (Read/Write)	Read / Link only	Read / Link only	Read / Attempt	Read / Attempt	Full Control
Institute Problems	Full Control	Full Control	Full Control	Denied	Read / Attempt (Own Inst)	Full Control
Global Contests	Full Control (Read/Write)	Read only	Read only	Read only	Compete	Compete
Institute Contests	Full Control	Full Control	Full Control	Denied	Compete (Own Inst)	Full Control
Study Materials	Global Materials	Local (Full) + Global (Read/Use)	Local (Full) + Global (Read/Use)	Local (Full) + Global (Read/Use)	Local (Full) + Global (Read/Use)	Local (Full) + Global (Read/Use)
Question Bank	Global Bank	Local (Full) + Global (Read/Use)	Local (Full) + Global (Read/Use)	Local (Full) + Global (Read/Use)	Local (Full) + Global (Read/Use)	Local (Full) + Global (Read/Use)
AI Viva Practice	Take Session (Global Bank)	Take Session (Global + Local Bank)	Take Session (Global + Local Bank)	Take Session (Global + Local Bank)	Take Session (Global + Local Bank)	Take Session (Global + Local Bank)

Feature Area	Super Admin (<code>ADMIN</code>)	Institute Admin (<code>INSTITUTE_ADMIN</code>)	Batch Manager (<code>BATCH_MANAGER</code>)	Mentor (<code>MENTOR</code>)	Global Student (<code>USER</code>)	Institute Student (<code>USER</code>)
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AI Viva Settings	🔧 Full Control (Read/Write)	❌ Denied	❌ Denied	❌ Denied	❌ Denied	❌ Denied
Global Problems	🔧 Full Control (Read/Write)	👁️ Read / Link only	👁️ Read / Link only	👁️ Read / Link only	👁️ Read/Attempt	👁️ Read/Attempt
Institute Problems	❌ Denied	🔧 Full Control	🔧 Full Control	🔧 Full Control	❌ Denied	👁️ Read/Attempt (Own Inst)
Global Contests	🔧 Full Control (Read/Write)	👁️ Read only	👁️ Read only	👁️ Read only	🏆 Compete	🏆 Compete
Institute Contests	❌ Denied	🔧 Full Control	🔧 Full Control	🔧 Full Control	❌ Denied	🏆 Compete (Own Inst)
Study Materials	🔧 Global Materials	🔧 Local (Full) + 👁️ Global (Read/Use)	🔧 Local (Full) + 👁️ Global (Read/Use)	🔧 Local (Full) + 👁️ Global (Read/Use)	❌ Denied (Staff Only)	❌ Denied (Staff Only)
Question Bank	🔧 Global Bank	🔧 Local (Full) + 👁️ Global (Read/Use)	🔧 Local (Full) + 👁️ Global (Read/Use)	🔧 Local (Full) + 👁️ Global (Read/Use)	❌ Denied (Staff Only)	❌ Denied (Staff Only)
AI Viva Practice	❌ Denied	❌ Denied	❌ Denied	❌ Denied	🎯 Take Session (Global Bank)	🎯 Take Session (Global + Local Bank)
Live Kit Streaming	👤 Host global/scoped streams	👤 Host streams in their inst	👤 Host streams in their inst	👤 Host streams in their inst	❌ Denied	👤 Join stream (Own Batch)

🎯 4. Detailed Flow of Core Modules

📁 Coding Problems & Contests

1. **Adding Problems:** When an Institute Admin/Mentor adds a problem, it automatically locks to their `instituteId`. Super Admin's problems are tagged `null` (Global). 2. **Display Switcher:** Institute staff see two tabs: "Your Institute" and "Global". Global tab actions hide edit/delete buttons for non-super-admins. 3. **Contest Linking:** Institute staff can link both Global and Institute problems to contests, but they can only publish target contests to cohorts belonging to their own institute.



AI Viva Question Pools

1. **Generation:** Mentors generate questions by uploading PDF study materials. If uploaded by a mentor, the document and all approved questions generated from it inherit the mentor's `instituteId`. 2. **Student Session Start:** When a student starts an AI Viva: * If **Institute Student**, the backend loads questions matching: $\text{where} = \{ \text{subject: chosen}, \text{OR: } [\{ \text{instituteId: null} \}, \{ \text{instituteId: studentInstId} \}] \}$ * If **Global Student**, the backend loads questions matching: $\text{where} = \{ \text{subject: chosen}, \text{instituteId: null} \}$